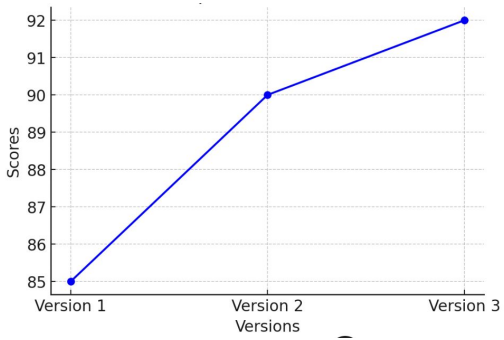
LLMs4review

- Task: LLMs on reviewing Paper on Arxiv v1/v2…
  - Could LLMs identify the correct version of paper acceptance(By score)
  - Is there a linear relationship with the LLM predicted score and the version number of Arxiv

[Submitted on 18 Nov 2024 (v1), last revised 19 Dec 2024 (this version, v2)]

[Submitted on 18 Nov 2024 (v1), last revised 3 Jun 2025 (this version, v3)]



Here is a detailed and critical review of the paper titled **"Quantifying Preferences of Vision-Language Models via Value Decomposition in Social Media Contexts"**, scored according to ACL-style criteria on a **0–5** scale.

Overall Score: 3.5 / 5

This paper presents an original benchmark and interesting empirical insights into VLM preference modeling, but suffers from significant methodological and clarity issues that limit its impact.

Here is a detailed and **critical ACL-style review** of the paper **"Value-Spectrum: Quantifying Preferences of Vision-Language Models via Value Decomposition in Social Media Contexts"** based on the ACL 0–5 scale:

Overall Score: 4 (Accept)

Recommendation: 3 (Weak Accept)

Justification:
This paper explores a **timely and under-explored topic**—evaluating VLMs via human value theory and persona simulation. Its methodological ambition and novel dataset are significant, but its conceptual clarity, statistical rigor, and theoretical grounding are not yet at the level of a top-tier ACL paper. With

Version 1 of our paper 3.5/5

Version 2 of our paper 4/5

Version 3 of our paper 3/5

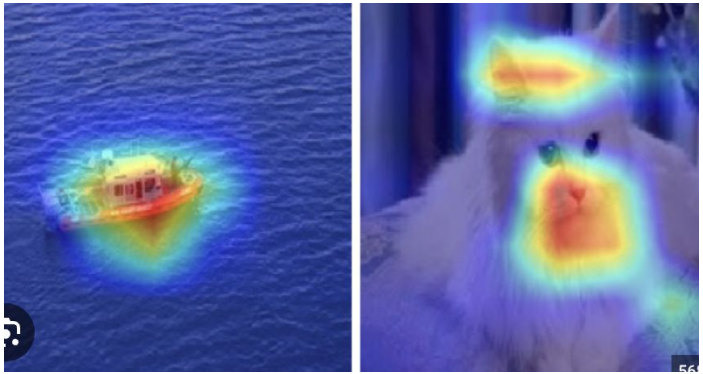
- Objective
  - Discover whether LLMs are consistent in paper reviewing
- Hypothesis
  - LLMs are good in establishing positive relationship
  - LLMs are not good in establishing linear relationship
    - Discovering the inconsistency

- Dataset
  - from papers(ICLR…) collect score of paper and their versions links
  - Feed to LLMs to get response for each version, obtain score
- Analysis
  - Discover linear relationship between paper version and Ai predicted scores
  - Discover Accuracy in predicting accepted versions

Title	Official Score	Official Acceptance Vers	Version1 Link	Version2 Link	Version3 Link	V1 Score	V2 Score	V3 Score Predicted	Acceptance Vers
Paper 1	1	Version1	http://example.com/pape	http://example.com/pape	http://example.com/pape	2.37	1.05	1.55	Version1
Paper 2	5	Version1	http://example.com/pape	http://example.com/pape	http://example.com/pape	4.93	3.76	4.59	Version1
Paper 3	4	Version1	http://example.com/pape	http://example.com/pape	http://example.com/pape	4.78	2.18	1.85	Version1

Technical Option I - Interpretability

- Using Attention Map to visualize attention on different paper version
  - See where the attention is at (abstract? Conclusion?)
  - Qualitative analysis on several cases
  - interpretable
- Only applicable on Transformer based model
  - Lamma 7b
- Usage
  - 4090/5090



Technical Option II: Fine Tuning

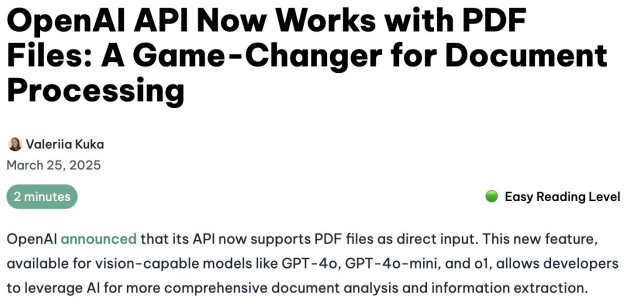
- Fine-tune a model using the version examples
  - Develop a better version that beats existing versions
  - Using human labeled dataset that marks different version and score
  - Use 4090 to finetune a model
  - Human evaluate the fine-tuned model performance
- Challenge
  - Need to compete with other models
  - https://arxiv.org/pdf/2503.08569
    DeepReviewer-14B
  - https://github.com/zhu-minjun/Researcher
    Cyclereview- 70B

Table 2: Results comparison between open-source LLMs and baselines on prediction tasks.

Model \ Metrics	Acceptance Prediction				Score Prediction	
	Accuracy	Precision	Recall	F1	MAE	MSE
LLaMA-3.1-8B (zero-shot)	60.19±0.00	60.19±0.00	100.00±0.00	75.13±0.00	1.961±0.055	5.488±0.287
Qwen2.5-7B (zero-shot)	60.19±0.00	60.19±0.00	100.00±0.00	75.13±0.00	2.043±0.000	5.756±0.000
LLaMA-3.1-8B (SFT)	63.23±0.33	**73.66**±0.64	63.74±0.72	68.35±0.12	1.141±0.030	2.290±0.108
Qwen2.5-7B (SFT)	62.01±0.00	72.25±0.01	62.18±0.00	66.83±0.00	1.182±0.000	2.374±0.000
SEA-E	66.24±1.42	66.69±0.68	**91.71**±0.82	**77.23**±0.74	1.157±0.044	2.304±0.176
LLaMA-OpenReviewer-8B	59.76±0.21	70.92±0.25	53.59±1.05	61.05±0.77	1.197±0.012	2.413±0.021
DeepReviewer-14B	**67.89**±1.99	65.75±1.31	84.43±1.87	73.93±1.51	**1.104**±0.021	**2.018**±0.087
CycleReviewer-8B	54.18±0.27	73.44±1.96	33.25±2.01	45.73±1.50	1.321±0.008	2.941±0.015

- Challenges
  - Data collection: Which version is the final version, which version is the accepted version
    - Human evaluation
  - Data Cleaning: Only want paper with several versions(>n) only want papers that are (ACCEPTED? NOT ACCEPTED? STratified?)
- Cost
  - GPT4o
  - Approximately $0.25 for a 20-page PDF using GPT-4o.
  - 3 versions X 0.25 X 1000 = 750 $

- To be decided
  - What paper, which range (ICLR?)
  - What topic (only Ai conference, physics? Math? Psychology?)
  - How PDF is presented -> only text? Directly?
  - Do we assume the final version is accepted?
  - How many papers do we want to do?
  - Do we want only accepted paper?



OpenAI API Now Works with PDF Files: A Game-Changer for Document Processing

Valeriia Kuka
March 25, 2025

2 minutes

Easy Reading Level

OpenAI announced that its API now supports PDF files as direct input. This new feature, available for vision-capable models like GPT-4o, GPT-4o-mini, and o1, allows developers to leverage AI for more comprehensive document analysis and information extraction.

具体是哪个journal

哪个dataset

不同会议标准不一样